

Predicting fine-scale lek site selection of greater sage-grouse (*Centrocercus urophasianus*) in Nevada, USA

Revised draft — 2026 analysis (v4 pipeline)

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Abstract

Following a significant population reduction and the degradation of sagebrush habitats throughout the Great Basin, greater sage-grouse (*Centrocercus urophasianus*) has become a species of conservation concern. Rangeland managers continue to seek ways to improve management strategies by accurately locating and characterizing habitat for sage-grouse, including lek site locations. Lek counts have proven useful for long-term population monitoring; understanding the selection of lek sites by sage-grouse remains a priority for conservation efforts. The purpose of this study is to locate areas with the highest potential of lek presence and identify environmental attributes characteristic of lek sites using a resource selection function (RSF). We incorporated 30-m resolution spatial data with the coordinate locations of currently active lek sites to model site selection across the species' 152,458-km² distribution in Nevada, USA. We fit fifteen candidate models to predict lek occurrence: twelve a priori covariate combinations and a planned second-stage decomposition distinguishing paved from unpaved road classes. Leks were more likely to occur at higher elevations (model-averaged $\beta = 0.94$, 85% CI 0.85 – 1.03) and less likely to occur on steeper slopes ($\beta = -1.31$, 85% CI -1.47 – -1.16), on rugged terrain measured by the Vector Ruggedness Measure ($\beta = -1.02$, 85% CI -1.27 – -0.76), and on level ground lacking measurable slope ($\beta = -1.20$). Grassland cover was favored over the shrubland reference ($\beta = 0.44$, 85% CI 0.15 – 0.72), while conifer cover held fewer than five of 718 lek sites and, pooled with other rarely used classes, was strongly avoided. Distinguishing road classes revealed opposing associations: leks sat farther from paved roads ($\beta = 0.10$, 85% CI 0.03 – 0.16) and, more weakly, closer to unpaved roads and vehicular trails ($\beta = -0.09$, 85% CI -0.19 – 0.01). The averaged model discriminated well (AUC = 0.81; spatially blocked cross-validation 0.80 ± 0.04) and, excepting an attenuated grassland effect, its predictions were robust to post-2018 vegetation change at lek sites. The analysis is implemented as a fully scripted, openly licensed pipeline in which every environmental input is fetched from its authoritative public source and every run is exactly reproducible. Our results are useful for rangeland managers and researchers in Nevada seeking to understand and improve greater sage-grouse population estimates using lek locations and counts.

Keywords: habitat selection, modeling, Nevada, resource selection function

Introduction

Greater sage-grouse (*Centrocercus urophasianus*; hereafter sage-grouse) is a sagebrush obligate bird species that relies on intact sagebrush ecosystems for reproduction and survival [1]. Sage-grouse populations have generally been in decline since the 1960s in response to the loss, fragmentation, and degradation of sagebrush habitat (WAFWA, 2015). Connelly and Braun (1997) [2] estimated that, by 1997, sage-grouse breeding populations had declined by 17–47% (range-wide average of 33%). Urban expansion, resource extraction, invasion of non-native exotic grasses, invasion of pinyon-juniper woodland, wildfire and novel disturbance regimes, and over-utilization of rangelands through excessive grazing have all contributed to the decline of sage-grouse [2, 5]. The potential consequence of these impacts is increased fragmentation and isolation of sage-grouse populations [6], and range-wide monitoring continues to document population concern, including in the Bi-State region of Nevada and California [36].

Sage-grouse are a lekking bird species: males gather each spring at traditional communal display grounds — leks — situated in open areas, and breeding is concentrated at these sites [1, 7]. Because sage-grouse congregate at lek sites during the breeding season, these areas are often used as a focal point for monitoring bird populations and establishing management priorities and conservation objectives [3, 8]. Lek count data can help determine the presence and/or abundance of nesting habitat across the surrounding landscape [6]. Active leks and peak counts of male sage-grouse have been linked to source habitats [9], suggesting that the density of sage-grouse at lekking areas is a reliable indicator of habitat quality [10]. Monitoring lek sites throughout the species range can help managers and researchers better understand and protect sage-grouse populations. Though some variation in methodology and sampling biases exists, lek monitoring is still useful for informing sage-grouse-related conservation decisions [11, 12], and lek-count time series now anchor range-wide early-warning systems for population declines [37]. A comprehensive knowledge of existing lek locations and site selection is critical for informed conservation and management decisions related to sage-grouse.

Our objectives were to 1) predict lek locations throughout the greater sage-grouse range in Nevada using landscape data from currently active lek sites and 2) describe the characteristics of active lek sites throughout their range in Nevada using a resource selection function. Active leks can be used to understand characteristics of potentially unknown active lek sites and provide a basis for new areas to survey for lek counts. Many studies have incorporated leks in their analyses and predictions of sage-grouse habitat use and quality in Nevada [8, 13, 14, 15, 16, 17, 18, 19], and recent range-wide products integrate space use, habitat selection, and survival at broad spatial scales to guide land-use planning [31]. However, these studies either delineated most or all landscape variables at relatively coarse resolutions (1-km²), incorporated lek location data from shorter time frames (5–6 years), did not predict new lek locations, or utilized location data other than lek sites (i.e., telemetry data). Our study improves upon past research as it incorporates fine-scale landscape variables (30-m resolution), builds upon a rich statewide dataset, and provides land managers with specific areas to target future lek surveys and conservation planning.

Methods

Study Area

We determined the extent of the study — **152,458 km²** — using the sage-grouse species distribution range for the state of Nevada, USA [32], combined with a 5-km buffer around active leks that fell outside the traditional range (Fig. 1). The range polygon is the current-range layer compiled for the U.S. Fish and Wildlife Service’s 2015 status review, which integrated the Schroeder et al. (2004) range delineation with the Priority Areas for Conservation identified in the 2013 Conservation Objectives Team report; it is distributed through USGS ScienceBase and was clipped to the Nevada state boundary for this analysis. Sage-grouse inhabit rangelands throughout the state generally characterized as high elevation, cool desert ecosystems with hot,

dry summers and cold winters. Wyoming big sagebrush (*Artemisia tridentata wyomingensis*), low sagebrush (*Artemisia arbuscula*), and black sagebrush (*Artemisia nova*) shrub species largely dominate the areas that fall below 2,100-m in elevation. At higher elevations, mountain big sagebrush (*Artemisia tridentata vaseyana*) is the dominant vegetation. The dominant conifers in this range include Utah juniper (*Juniperus osteosperma*) and single-leaf pinyon pine (*Pinus monophylla*) that tend to colonize mid- to high-elevation and mesic sagebrush habitats (Coates et al., 2020). Dominant grasses include cheatgrass (*Bromus tectorum*), bottlebrush squirreltail (*Elymus elymoides*), crested wheatgrass (*Agropyron cristatum*), Sandberg bluegrass (*Poa secunda*), and many other annual and perennial species.

Study area: sage-grouse range in Nevada
(USFWS 2015 range; 152,458 km² with lek buffers)



Figure 1. Study area highlighting the distribution of the greater sage-grouse in the state of Nevada, USA: the distribution range and 5-km buffers around known active lek sites used in this analysis.

Data Acquisition

To determine lek locations, we acquired the state of Nevada's historical lek count dataset from the Nevada Department of Wildlife (NDOW) in 2020. The dataset included lek center GPS locations and annual male counts for all known leks throughout the state of Nevada from 1947–2018. Lek sites were characterized as active, inactive, historic, no activity, or unknown. This study analyzed leks that were categorized as “active” as of 2018 and located within Nevada ($n = 718$, from 2,253 records statewide), omitting all others. (An earlier draft

reported $n = 716$; the register contains 718 in-state active records, and the full pipeline reconciles the count programmatically.)

Environmental and anthropogenic data were acquired programmatically from their authoritative public sources, and every acquisition writes a provenance manifest (source URL, publication date, size) alongside the data.

Elevation comes from the USGS 3D Elevation Program (3DEP), the national program that is systematically replacing legacy National Elevation Dataset sources with lidar-derived elevation. We used the seamless 1 arc-second (~30-m) product, distributed as one-degree cloud-optimized GeoTIFF tiles through The National Map API. Because the API returns multiple product vintages for the same tile footprint, the fetch step deduplicates to the most recent publication per cell; the 72 tiles used here carry publication vintages spanning 2013–2026, reflecting the progress of lidar acquisition across Nevada.

Vegetation comes from LANDFIRE, the interagency (U.S. Department of the Interior and USDA Forest Service) landscape-scale mapping program. We used the Existing Vegetation Type (EVT) layer from the LF2025 release at its native 30-m resolution, requested as a statewide clip through the LANDFIRE Product Service API, and classified sites by the layer’s physiognomy attribute (EVT_PHYS), which groups vegetation by growth form (e.g., shrubland, grassland, conifer) rather than by floristic composition.

Roads come from the U.S. Census Bureau’s TIGER/Line extracts (2024 vintage) of the MAF/TIGER database — the successor to the Feature Class Code (FCC) roads data used in earlier analyses of this system, now classified under the MAF/TIGER Feature Class Code (MTFCC) schema. County-level road files include every class from primary highways (S1100) through local roads (S1400) and unimproved vehicular trails (S1500); all 17 Nevada county files were merged (202,967 segments).

External validation uses the USGS data release accompanying Open-File Report 2024-1018 [31]: telemetry-based habitat-selection and survival rasters for greater sage-grouse in Nevada and northeastern California (ver. 3.0, September 2025) [40], from which we used the spring (breeding season) habitat-selection categories.

Input	Service	Product	Volume
Elevation	USGS The National Map API (tnmaccess.nationalmap.gov/)	3DEP 1 arc-second, cloud-optimized GeoTIFF tiles	72 one-degree tiles, 3.6 GB, tile vintages 2013–2026, deduplicated to the most recent publication per cell
Vegetation	LANDFIRE Product Service API (lfps.usgs.gov)	LF2025 Existing Vegetation Type, EVT_PHYS attribute	statewide clip, 30 m
Roads	U.S. Census TIGER/Line (2024)	all-classes county road files (MTFCC schema)	17 counties, 202,967 road segments
Range	USGS ScienceBase API	USFWS 2015 Status Review current range [32]	556 polygons, clipped to Nevada

The DEM tiles were mosaicked and warped to a common 30-m analysis grid (EPSG:5070, Albers equal-area), from which all terrain derivatives were computed; the vegetation and distance-to-road surfaces were aligned to the same grid. Intermediate layers are staged as cloud-optimized GeoTIFF and GeoParquet.

Statistical Analysis

We conducted a resource selection function (RSF) based on 718 known lek locations throughout the state of Nevada to 1) predict areas with potential lekking activity and 2) quantify active lek site characteristics. We used a use-availability study design where each lek site was considered a sampling unit [20]. The leks were considered the “used” points in the analysis and were compared to randomly generated “available” points. We generated ten available points per used lek ($n = 7,180$) as a seeded uniform random sample within the study area, making the availability draw exactly reproducible.

Predictor Variables

Generating RSF predictions requires measurements of multiple predictor variables at each “used” and “available” site. Environmental data included LANDFIRE vegetation cover types (Existing Vegetation Type physiognomy) [28] and elevation from a 30-meter digital elevation model (DEM). The vegetation cover layer’s physiognomy classes include conifer, conifer-hardwood, exotic herbaceous, exotic tree-shrub, grassland, quarries-strip mines-gravel pits-well/wind pads, riparian, shrubland, sparsely vegetated, developed, agricultural, and open water. Classes with fewer than 10 sites overall, or fewer than 5 used leks, were pooled into a single “other” category: a class containing almost no used points cannot support a stable coefficient, and retaining such classes produced quasi-complete separation (inflated coefficients with standard errors in the hundreds) in earlier fits of this dataset. DEM derivatives included aspect, slope, curvature, and ruggedness (Fig. 2). Curvature is total surface curvature — the sum of profile and tangential components, in the sign convention of the ArcGIS Curvature tool — where positive values indicate locally convex-upward terrain (ridges, knolls) and negative values indicate concave terrain (swales, drainages). We elected to use a 30-m DEM to examine patterns at a fine scale across the region. We accounted for anthropogenic activity by calculating the Euclidean distance from each site to the nearest road. All data processing is automated and documented in an openly licensed pipeline built on GDAL, Python, and R; intermediate layers are staged as cloud-optimized GeoTIFF and GeoParquet.

We categorized aspect into eight cardinal directions — north (N), east (E), south (S), west (W), northeast (NE), northwest (NW), southeast (SE), and southwest (SW) — plus a ninth category, Flat, for cells with slope below 0.5° , where aspect is undefined; conflating such cells with north-facing terrain (the default behavior of common aspect tools) would contaminate the aspect effects, because leks preferentially occupy gentle ground. Ruggedness was quantified with the Vector Ruggedness Measure (VRM) [33], computed over a 3×3 ($90\text{-m} \times 90\text{-m}$) window from the slope and aspect surfaces. VRM measures the dispersion of surface-normal vectors — a geometric property of terrain — and is decorrelated from slope by construction ($r = 0.40$ in our design), making it well suited to models that include both terms; it is the standard ruggedness metric in wildlife habitat analysis [33]. (Earlier analyses of this dataset used a range-position focal index whose values proved unstable across DEM generations and which entered those models as an effectively binary covariate; ruggedness results are therefore not directly comparable across versions of this analysis.)

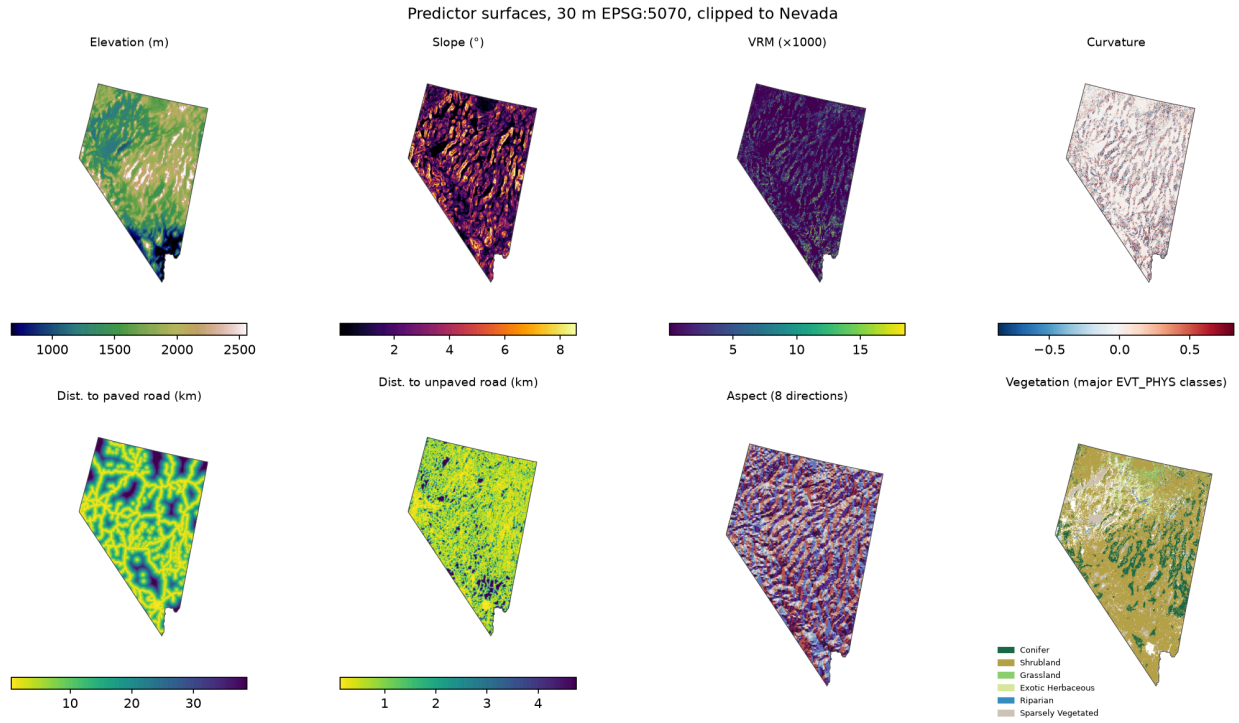


Figure 2. Predictor surfaces on the 30-m analysis grid, clipped to Nevada: elevation, slope, VRM, curvature, class-specific road distances, aspect, and major vegetation classes.

The roads dataset comprised all Census TIGER/Line road classes (MTFCC schema) — 100,790 km of road within the study area — dominated by local roads and unimproved vehicular trails (Table 1, Table 2; Figs. 3–4).

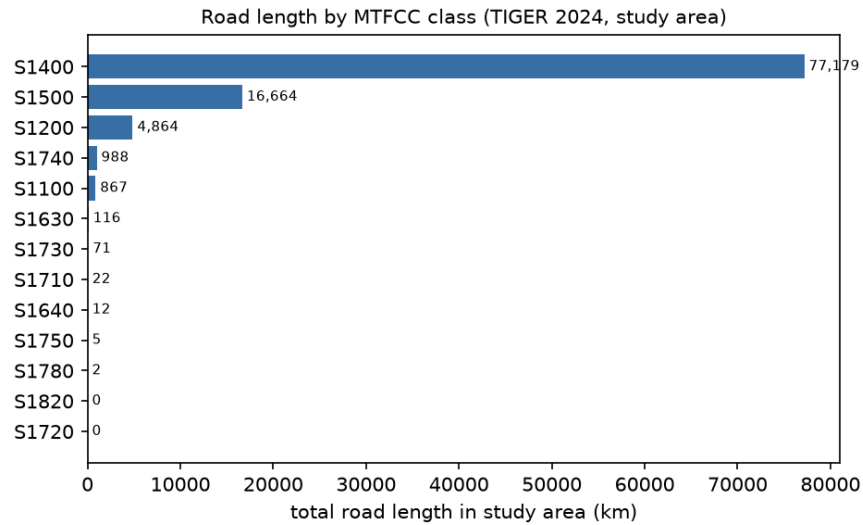


Figure 3. Road length in the study area by MTFCC class.

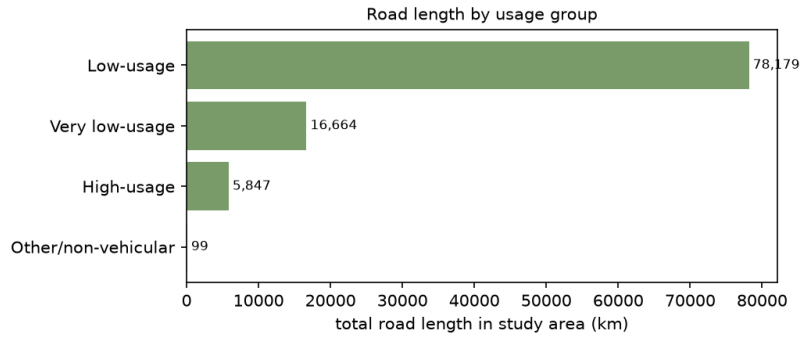


Figure 4. Road length in the study area by usage group.

Table 1. MTFCC classes present in the study area.

MTFCC	Description	Length (km)
S1400	Local neighborhood road, rural road, city street	77,179
S1500	Vehicular trail (4WD, unimproved)	16,664
S1200	Secondary road (state/county highway)	4,864
S1740	Private road for service vehicles	988
S1100	Primary road (interstate/major highway)	867
S1630	Ramp	116
S1730	Alley	71
S1710	Walkway/pedestrian trail	22
S1640	Service drive	12
S1750	Internal U.S. Census Bureau use	5
S1780	Parking lot road	2
S1820	Bike path or trail	0
S1720	Stairway	0

Table 2. MTFCC classes aggregated to usage groups.

Usage group	MTFCC classes	Length (km)
Low-usage	S1400, S1640, S1740	78,179
Very low-usage	S1500	16,664
High-usage	S1100, S1200, S1630	5,847
Other/non-vehicular	all others	99

From the merged network we computed three Euclidean-distance surfaces on the DEM grid, ensuring cell-location continuity with the elevation-derived rasters: distance to any road, distance to paved roads (MTFCC S1100, S1200, S1630), and distance to unpaved roads and vehicular trails (S1400, S1500). The class-specific surfaces allow the models to separate a disturbance signal (proximity to high-use roads) from an association with the unimproved routes that thread open terrain. Finally, elevation, aspect, slope, curvature, ruggedness,

the three road-distance measures, and vegetation cover types were sampled at each used and available location. To validate that we had enough random points to assess available habitat, we compared the mean of the study area’s pixels for each continuous variable (estimated with a dense 100,000-point Monte Carlo sample at full raster resolution) against the available-sample mean and its 85% confidence interval (S1 Table).

Resource Selection Function

We calculated the prediction values for the RSF using logistic regression with used ($n = 718$) and available ($n = 7,180$) points. Prior to running the models, we checked for multicollinearity among explanatory variables: no pair of covariates appearing together in any single model exceeded $|r| = 0.6$ (Fig. 5). (Distance to any road and distance to unpaved roads are nearly collinear, $r = 0.99$, because unpaved routes dominate the network; these two variables never co-occur in a model.) We fit the twelve a priori models of the original study design — combinations of the terrain, vegetation, and combined-roads covariates (Table 3). Because a single all-roads distance conflates road classes with opposing expected effects, we then constructed three additional models in a planned second stage, distinguishing paved (TIGER MTFCC S1100, S1200, S1630) from unpaved (S1400 local roads and S1500 vehicular trails) distances, to separate a disturbance hypothesis (leks avoid high-use roads) from an openness hypothesis (unimproved routes co-occur with the open terrain leks favor). All fifteen models entered a single ranking by Akaike’s information criterion corrected for small sample size (AICc); we note the road-class models were specified after the combined-roads models had been examined, and their support should be read in that light. In R [29], we fit each model with the base `glm` function and ranked models by AICc using the MuMIn package [34]. We tested our top three models’ variance inflation factors (VIF) to further inspect for multicollinearity, considering $VIF > 10$ indicative of collinearity [22, 27]; all variables in all top models had $VIF \leq 1.8$. To account for the uncertainty inherent in model selection, we averaged beta coefficients from our top three models using the full average in MuMIn, in which a covariate absent from a model contributes zero to that model’s term rather than being omitted, and report 85% confidence intervals, the interval width consistent with AIC-based model selection [43] (Table 4). Model discrimination was assessed with the area under the receiver operating characteristic (ROC) curve (AUC) — the probability that a randomly chosen used site receives a higher predicted value than a randomly chosen available site, where 0.5 indicates discrimination no better than chance and 1 indicates perfect ranking — on the fitted model average, under random 5-fold cross-validation, and under spatially blocked 5-fold cross-validation (k-means coordinate blocks), and calibration was inspected by deciles of predicted probability. The final surface was evaluated with the continuous Boyce index [41, 42], a presence-only measure appropriate to use-availability designs: the range of predicted probabilities is partitioned into classes, and each class’s predicted-to-expected (P/E) ratio is computed as the proportion of used leks falling in the class divided by the proportion of the study area the class occupies. The index is the Spearman rank correlation between P/E and class midpoint and ranges from -1 to 1 ; values near 1 indicate lek density rises monotonically with predicted suitability, 0 indicates a surface no better than chance, and negative values indicate counter-prediction. The surface was also externally compared against the U.S. Geological Survey’s independent, telemetry-based spring habitat selection categories for Nevada and northeastern California (ver. 3.0, 2025) [40] at 150,000 seeded random locations. That product classes the landscape into four percentile thresholds of a habitat-selection index — non-habitat (below the 5th percentile), low (5th–25th), moderate (25th–50th), and high (above the 50th) — classes of deliberately unequal area, so we compare against our five equal-area quantile bins ordinally (rank correlation and a row-normalized cross-tabulation) rather than class-for-class.

To generate prediction values across the study area, the averaged model was applied directly to the covariate rasters: every 30-m cell in the study area was scored in blocked array computation, producing a probability surface that was then classified into five bins of use probability using a quantile classification. Because the used:available ratio is fixed by design, fitted values are relative indices of selection rather than absolute probabilities of lek occurrence [41]; we accordingly interpret the surface through its ranks — quantile classes,

rank correlations, and the Boyce index — rather than through the magnitude of any single predicted value. To assess sensitivity to post-2018 vegetation change (including large wildfires such as the 2018 Martin Fire), we cross-tabulated each used lek's vegetation class between the LF2016 and LF2025 LANDFIRE eras and refit the top model excluding all class-changed leks.

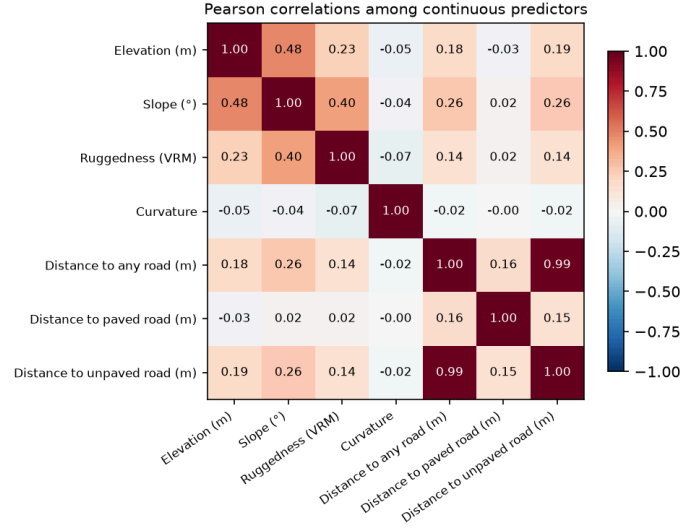


Figure 5. Pearson correlations among continuous predictors.

Results

Our statewide results are based on 718 active lek sites and 7,180 randomly generated available points. Used sites separated visibly from availability on slope, ruggedness, and elevation before any model was fit (Figs. 6–7).

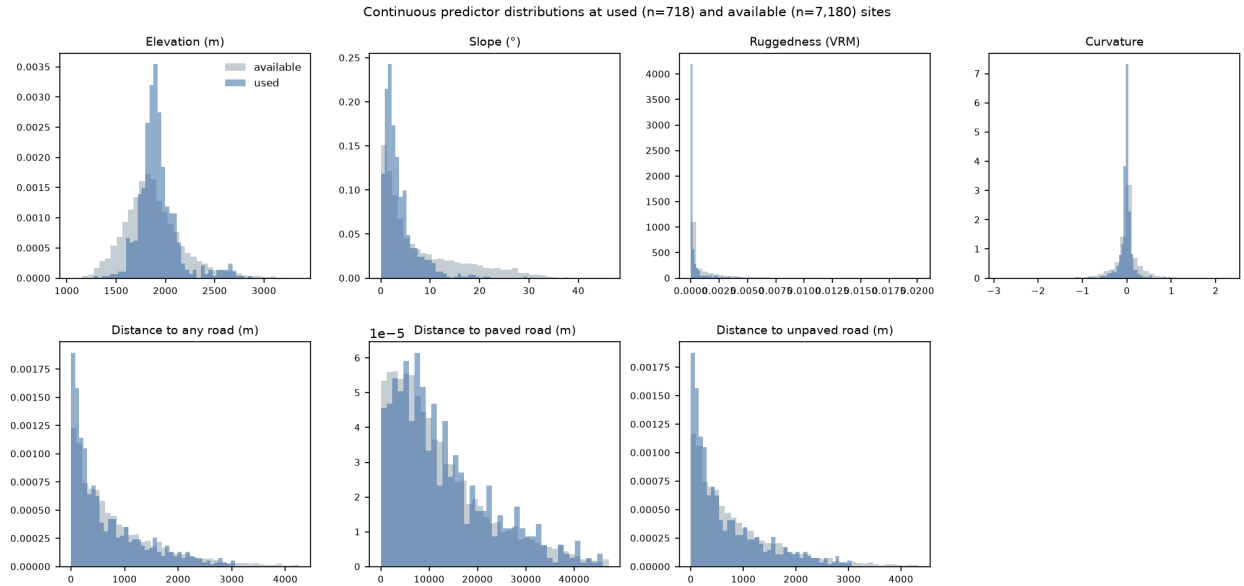


Figure 6. Continuous predictor distributions at used and available sites.

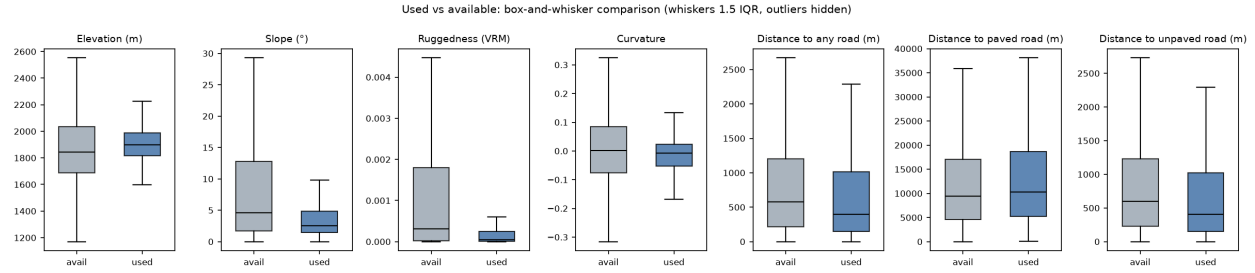


Figure 7. Used versus available sites, box-and-whisker (whiskers 1.5×IQR, outliers suppressed for scale).

The three road-class models ranked above all twelve original formulations (Table 3): distinguishing paved from unpaved road proximity improved every model it entered. Ruggedness, slope, direction (aspect), elevation, curvature, and vegetation cover types occurred in all top three models. VIF for the top three models indicated limited correlation among explanatory variables (1.0–1.8).

Table 3. Model selection for the fifteen candidate models (twelve a priori formulations, m1–m12; three second-stage road-class models, m13–m15).

Model	Terms	df	log-lik	AICc	delta	weight
m13	paved + unpaved + curvature + ruggedness + slope + direction + elevation + vegetation	20	-1,996	4,032	0	0.6693
m14	paved + curvature + ruggedness + slope + direction + elevation + vegetation	19	-1,998	4,034	2.208	0.2219
m15	unpaved + curvature + ruggedness + slope + direction + elevation + vegetation	19	-1,999	4,036	4.515	0.07001
m1	roads + curvature + ruggedness + slope + direction + elevation + vegetation	19	-2,000	4,038	5.697	0.03877
m2	roads + ruggedness + slope + direction + elevation + vegetation	18	-2,012	4,060	28.557e-07	
m7	ruggedness + slope + direction + elevation + vegetation	17	-2,013	4,060	28.05436e-07	
m8	roads + slope + direction + elevation + vegetation	17	-2,031	4,096	63.62027e-14	
m5	curvature + slope + elevation + vegetation	9	-2,065	4,149	116.2794e-26	
m10	slope + elevation + vegetation	8	-2,069	4,155	122.8461e-27	
m4	roads + ruggedness + direction + elevation + vegetation	17	-2,120	4,273	241.3725e-53	
m11	ruggedness + slope + vegetation	8	-2,174	4,365	332.3816e-73	
m3	roads + ruggedness + slope + vegetation	9	-2,174	4,366	334.2838e-73	
m9	roads + curvature + slope + vegetation	9	-2,180	4,379	346.6576e-76	
m6	roads + slope + vegetation	8	-2,187	4,389	357.1925e-78	
m12	roads + direction + elevation + vegetation	16	-2,223	4,477	445.2442e-97	

Table 4. Model-averaged covariate effects (top three models, 85% CIs; vegetation contrasts relative to shrub-land).

Term	Estimate	85% CI low	85% CI high
(Intercept)	-2.521	-2.698	-2.344
scale_PavedProximity	0.09591	0.02718	0.1646
scale_UnpavedProximity	-0.08726	-0.1871	0.01261
scale_Curvature	-0.4219	-0.5506	-0.2932
scale_Ruggedness	-1.015	-1.275	-0.7559
scale_Slope	-1.313	-1.466	-1.16
DirectionFlat	-1.196	-1.485	-0.908
DirectionN	-0.1003	-0.3369	0.1364
DirectionNE	-0.298	-0.5386	-0.05738
DirectionNW	-0.6373	-0.8713	-0.4033
DirectionS	-0.1135	-0.3504	0.1233
DirectionSE	0.01996	-0.1793	0.2192
DirectionSW	-0.6231	-0.9031	-0.3431
DirectionW	-0.6312	-0.8664	-0.3959
scale_Elevation	0.9394	0.8506	1.028
VegetationExotic Herbaceous	0.2039	-0.05506	0.4628
VegetationExotic Tree-Shrub	-0.5777	-1.06	-0.09563
VegetationGrassland	0.4354	0.1547	0.716
VegetationOther	-2.567	-3.062	-2.073
VegetationRiparian	-0.7259	-1.216	-0.2354

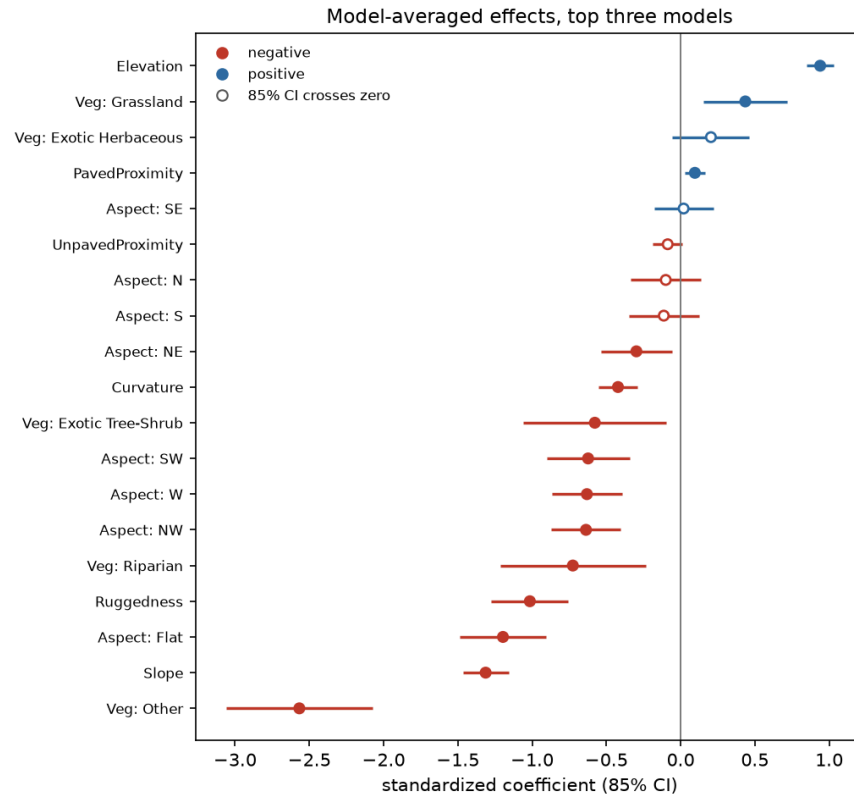


Figure 8. Model-averaged coefficients with 85% CIs (forest plot). Red indicates negative estimates and blue positive; open symbols mark effects whose 85% CI crosses zero. Vegetation (Veg:) rows are contrasts relative to shrubland; aspect rows are contrasts relative to east-facing slopes.

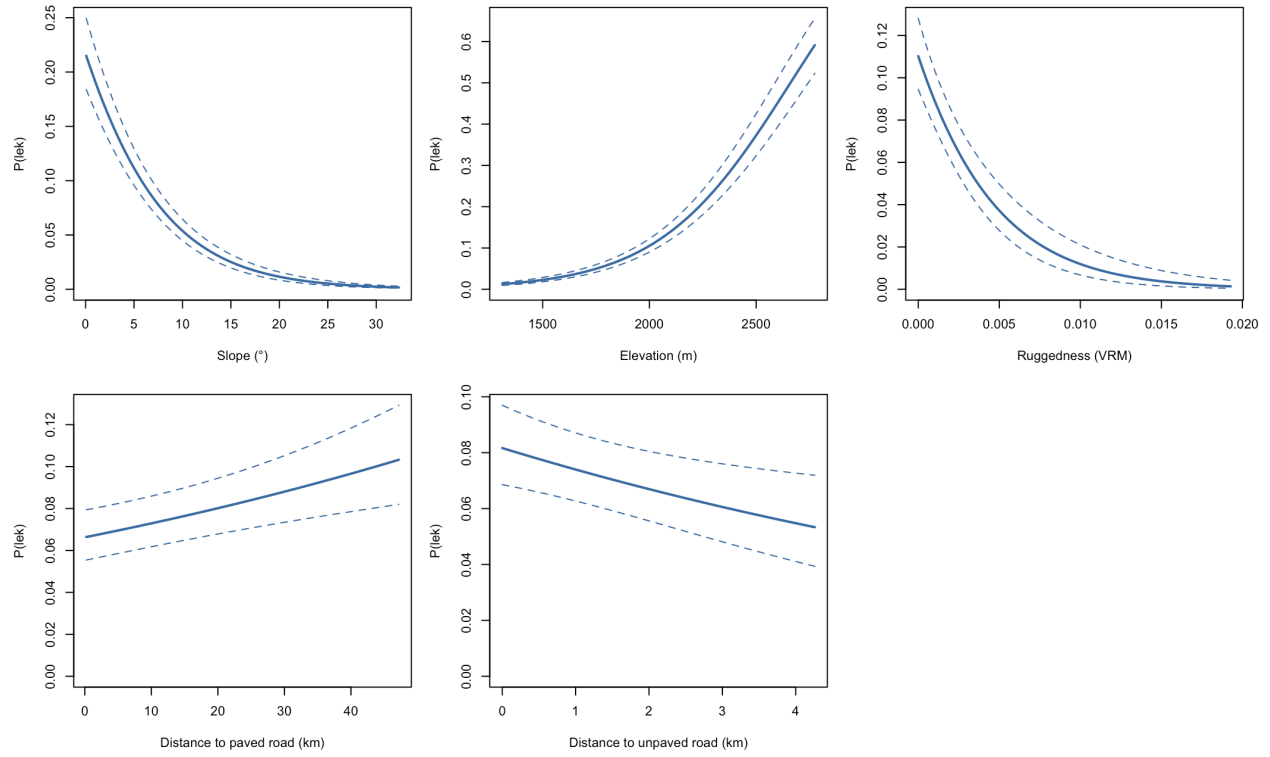
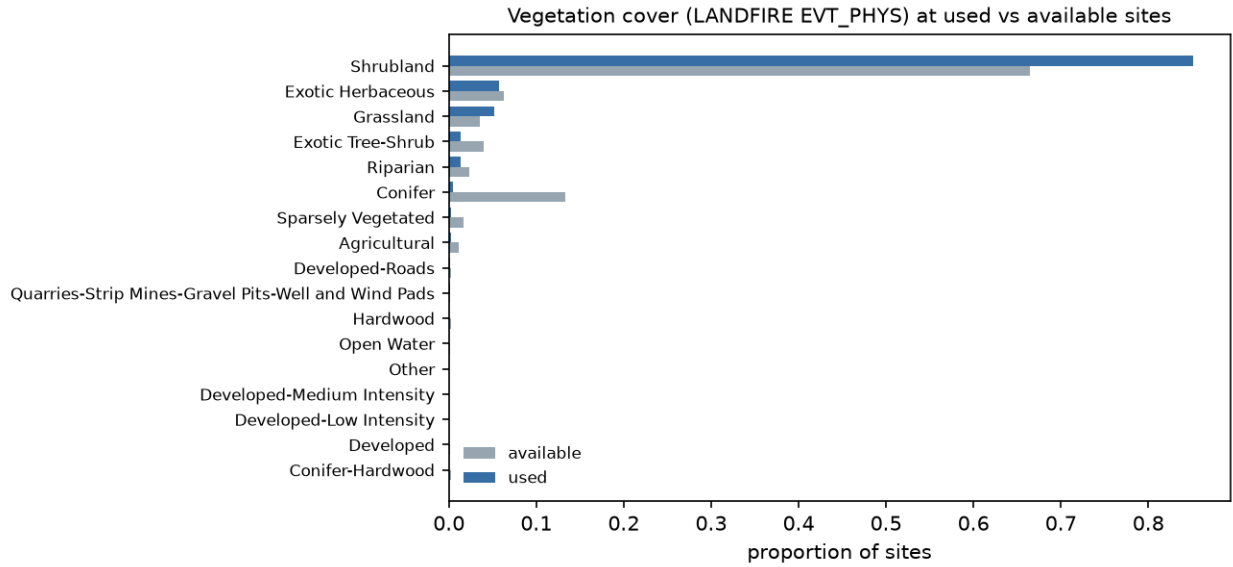


Figure 9. Marginal response curves from the top model: predicted probability of lek occurrence across each covariate's observed range, others held at reference values.

Based on the model-averaged results (Table 4; Figs. 8–9), leks were more likely to occur at higher elevations ($\beta = 0.94$, 85% CI 0.85 – 1.03) and less likely to occur on steeper slopes ($\beta = -1.31$, 85% CI -1.47 – -1.16) and on rugged terrain (VRM; $\beta = -1.02$, 85% CI -1.27 – -0.76); mean VRM at used lek sites was 0.00046 against 0.00197 across availability. Level ground lacking measurable slope – playa and lake-bed surfaces – was also strongly avoided (Flat; $\beta = -1.20$, 85% CI -1.48 – -0.91): leks occupy gentle, but not featureless, terrain. Relative to shrubland, grassland cover was selected ($\beta = 0.44$, 85% CI 0.15 – 0.72); exotic herbaceous cover trended positive without reaching significance ($\beta = 0.20$, 85% CI -0.06 – 0.46); exotic tree-shrub and riparian classes were avoided; and conifer cover contained fewer than five of the 718 used leks – near-complete absence – entering the pooled rarely-used category, which was strongly avoided ($\beta = -2.57$, 85% CI -3.06 – -2.07) (Fig. 10).



Leks were less likely to be located on NE-, NW-, SW-, and W-facing slopes relative to E-facing slopes; effects for N- and S-facing slopes did not differ from the reference at the 85% level. Road classes showed opposing associations: leks sat farther from paved roads than availability ($\beta = 0.10$, 85% CI 0.03 – 0.16), while the association with unpaved roads and vehicular trails was negative but did not exclude zero ($\beta = -0.09$, 85% CI $-0.19 - 0.01$; lek sites averaged 711 m from the nearest unpaved route). Curvature carried a significant negative coefficient ($\beta = -0.42$, 85% CI $-0.55 - -0.29$), indicating avoidance of locally convex microtopography.

The averaged model discriminated well between used and available sites (AUC = 0.81, Fig. 11; random 5-fold cross-validation 0.80 ± 0.01 ; spatially blocked 5-fold cross-validation of the top model 0.80 ± 0.04), and predicted values tracked observed use rates across deciles (Fig. 12) — internally consistent, noting that fitted values are relative selection indices rather than absolute occurrence probabilities. Evaluated against the final prediction surface, the continuous Boyce index was 0.92 (Fig. 13; S2 Table): lek density rises nearly monotonically across the predicted-suitability gradient, and cells in the highest probability class hold leks at more than six times the rate their area alone would predict. Externally, the surface correlated positively with the USGS telemetry-based spring habitat selection categories (Fig. 14; Spearman $\rho = 0.53$ at ~147,000 common locations): 46% of the cells the USGS classes as non-habitat fell in our lowest quantile bin and 41% of their high-habitat cells in our highest — aligned with independent data while retaining the complementary, lek-specific information of this analysis. In the vegetation-change sensitivity analysis, 35 of 718 used leks (4.9%) changed physiognomy class between LANDFIRE eras (8 showing the shrubland-to-herbaceous transition characteristic of fire); excluding them left terrain, road, and exotic herbaceous coefficients essentially unchanged, while the grassland coefficient attenuated from 0.43 to 0.21, and we interpret grassland selection with corresponding caution.

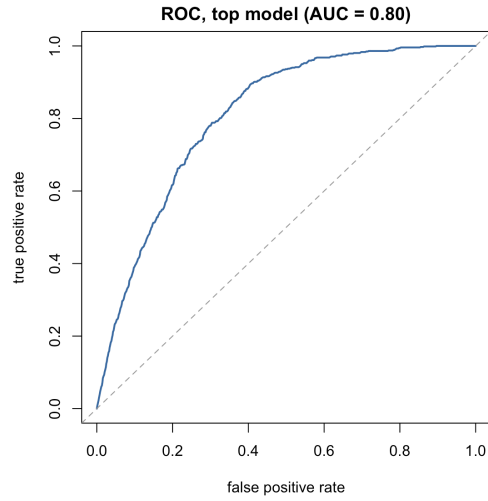


Figure 11. Receiver operating characteristic (ROC) curve for the top model.

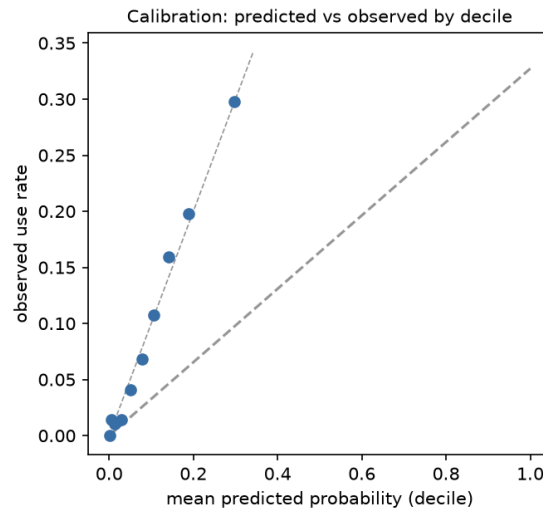


Figure 12. Calibration: mean predicted probability versus observed use rate by decile.

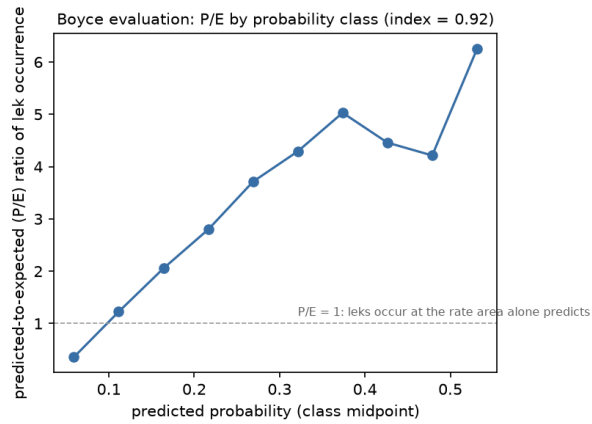


Figure 13. Boyce evaluation: predicted-to-expected ratio by probability class.

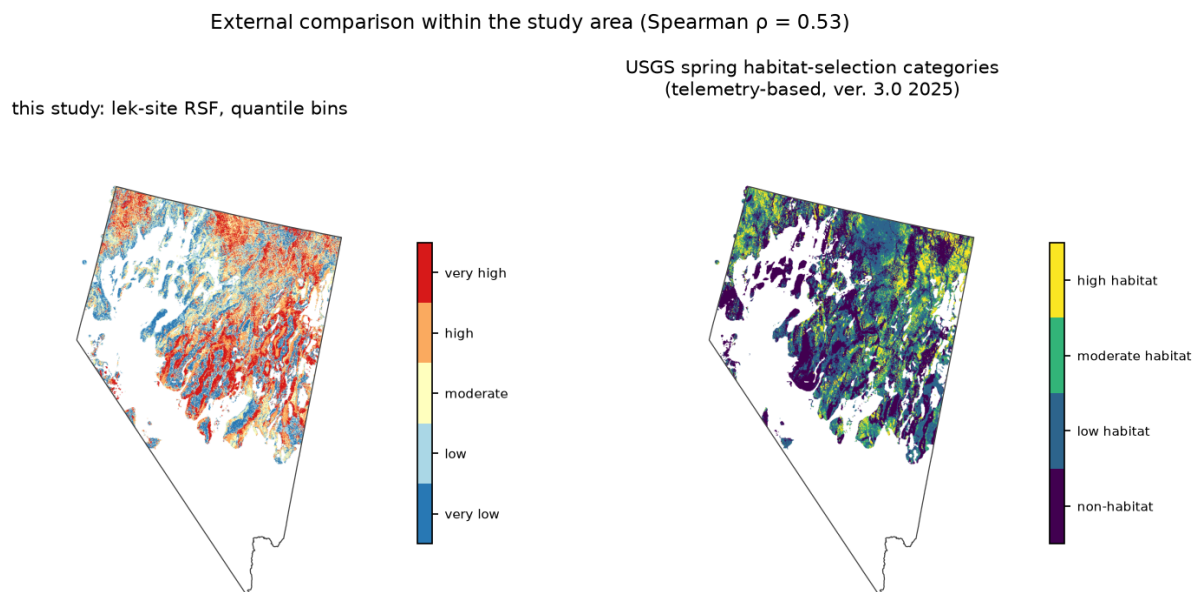


Figure 14. External comparison against the USGS telemetry-based spring habitat-selection categories, clipped to the study area.

Cross-tabulation (rows: USGS selection category, row-normalized; cell shading scales with each row's share): the gradient is monotone — 46% of USGS non-habitat falls in our lowest bin, and 41% of their high-habitat cells fall in our highest.

	ours very low	ours low	ours moderate	ours high	ours very high
USGS non-habitat	46%	29%	14%	7%	4%
USGS low habitat	9%	18%	25%	26%	22%
USGS moderate habi- tation	4%	13%	22%	27%	33%
USGS high habitat	5%	11%	18%	25%	41%

USGS classes are percentile thresholds of their habitat-selection index (non-habitat < 5th percentile; low 5th–25th; moderate 25th–50th; high > 50th) — four classes of deliberately unequal area, unlike our five equal-area quantile bins, so the comparison is ordinal rather than class-for-class. Because our bins are equal-area, every cell would read ~20% if the two maps were unrelated; departures from 20% are the signal.

Predicted probability at used leks was uncorrelated with peak male counts (Spearman $\rho = 0.04$), indicating the surface characterizes lek placement rather than lek size [6]. The classified surface is mapped statewide in Fig. 15, with each quantile class isolated in Fig. 16 and a 60-km window resolved at full 30-m detail in Fig. 17.

Predicted probability of greater sage-grouse lek occurrence
(quantile classification, five bins)

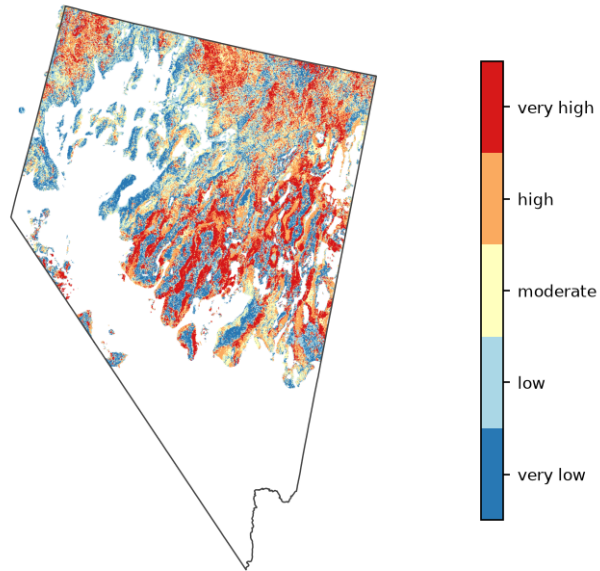


Figure 15. Predicted probabilities of greater sage-grouse lek occurrence in the state of Nevada, USA. Probabilities are classified into five bins of use probability using a quantile classification.

Bin	Share of study area	Probability range
very low	20.0%	< 0.006
low	20.0%	0.006–0.030
moderate	20.0%	0.030–0.078
high	20.0%	0.078–0.145
very high	20.0%	> 0.145

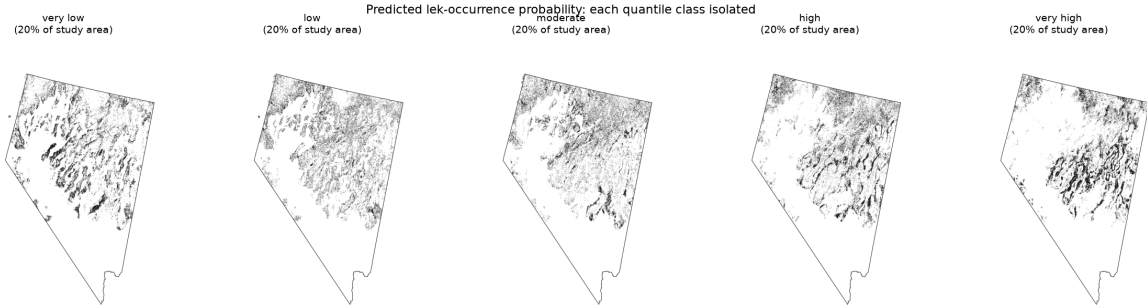


Figure 16. Each quantile class isolated (equal-area by construction). The highest-probability class alone constitutes the survey-targeting product: one-fifth of the study area capturing the terrain-and-vegetation conditions of active leks.

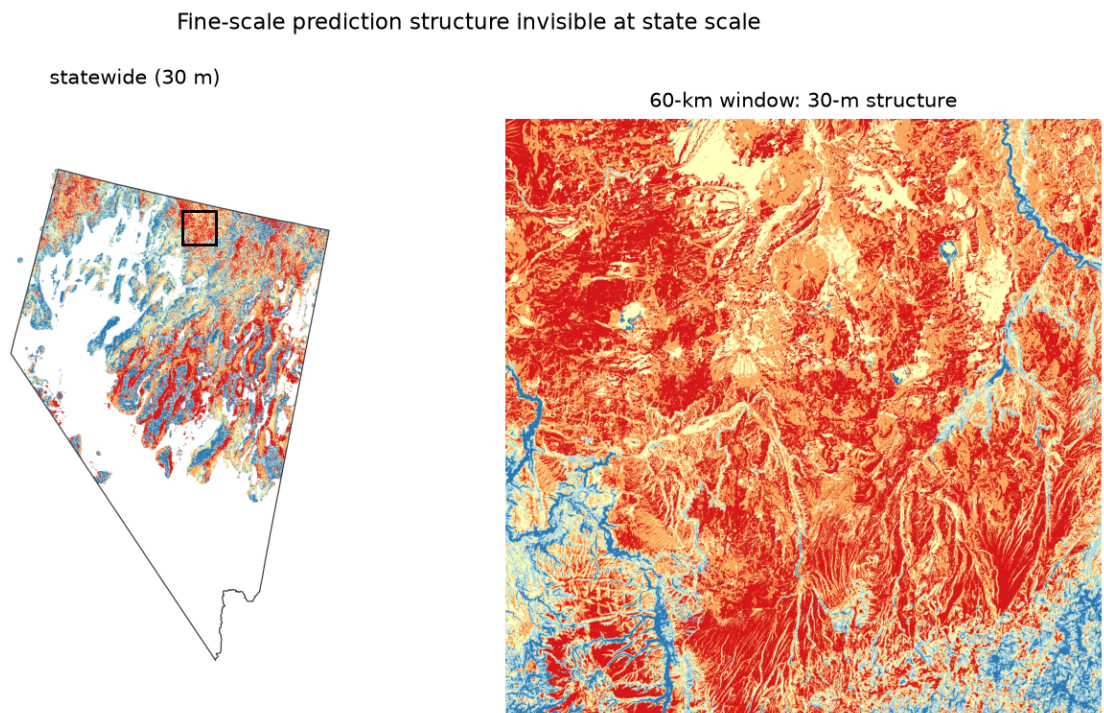


Figure 17. Fine-scale structure of the prediction surface: the statewide map with a 60-km window (Tuscarora Mountains area, selected for terrain diversity) resolved at full 30-m detail.

Discussion

Using fine spatial resolution information, our study builds upon a statewide data set providing land managers specific information to help identify and locate lek sites. This can help direct search effort for locating leks, and potentially better direct conservation efforts toward improving surrounding nesting cover that is critical for sage-grouse nesting success. It is well known that sage-grouse are highly associated with sagebrush for many life history stages [13, 14, 15, 16, 17, 18, 19]. We found grassland cover favored over the shrubland reference at lek sites (with exotic herbaceous trending the same direction), consistent with the open character of display areas [1, 7]; vegetation patterns that influence nesting occur at much broader scales, and males primarily select display areas in response to hens' preference for proximal nesting habitat. Two cautions attach to the

vegetation results. First, lek status is current to 2018 while the vegetation layer is LF2025; post-2018 disturbance, including Nevada's largest recorded wildfire (the 2018 Martin Fire), reclassified vegetation at 4.9% of used leks. Our sensitivity analysis excluding those leks left most coefficients unchanged but halved the grassland effect, indicating part of the grassland signal reflects post-2018 vegetation change at lek sites rather than selection. Second, conifer's near-complete absence at leks (fewer than five of 718 sites) precludes a stable class-specific coefficient; the avoidance is expressed through the pooled rarely-used category.

We found strong avoidance of conifer cover at lek sites, consistent with lek persistence declining as conifer encroaches on surrounding sagebrush [38]. This aligns with the active management literature on pinyon-juniper expansion: sage-grouse respond favorably to conifer removal, reoccupying restored nesting habitat near treatment areas [39], making lek-scale conifer avoidance directly actionable for treatment prioritization.

Leks were less likely to occur on steeper slopes and on rugged terrain, and more likely at higher elevations. Earlier work in Nevada reported mixed relationships among slope, elevation, and ruggedness [13, 14, 16, 18], and Coates et al. (2020) [18] found sage-grouse avoided rugged terrain in all seasons. Our results, measured with a geometric ruggedness instrument (VRM), are consistent with that avoidance and with the most recent work in a congeneric lekking system: Lautenbach et al. (2025) [35] found sharp-tailed grouse leks located in less rugged terrain with greater herbaceous cover, and proposed mechanisms — female detection of displaying males and sound propagation across smooth terrain — that apply equally to sage-grouse display ecology. Together these results suggest the smoothness of the display surface itself, distinct from slope, is a general feature of lek placement in prairie and shrub-steppe grouse.

Distinguishing road classes resolved what a single all-roads distance obscured: leks avoid proximity to paved, higher-use roads while sitting marginally closer to unpaved routes and vehicular trails. This is consistent with lekking activity favoring open, accessible sites [1, 7] and with the observation that unpaved, remotely located roads can host successful lek sites while busier, noisier roads may be inhospitable to lekking activity [13]. The class-split models outranked every single-distance formulation, suggesting that road class carries more ecological information for lek placement than road presence alone; the unpaved association, however, was weak and should not be over-interpreted.

We found that leks were less likely to be located on NE-, NW-, SW-, and W-facing slopes, and on flat aspectless ground, compared to E-facing slopes. Leks are typically found on less inclined surfaces, which would suggest that aspect would have minimal effect on lek location; however, easterly sites could have higher soil moisture conditions that would potentially influence surrounding plant community structure important for sage-grouse habitat. Precipitation was not a covariate included in our analysis; we recommend future studies incorporate climatic variables to further examine this relationship.

A vital consideration for managers tasked with conserving habitat for a species that exhibits seasonal variation in habitat selection is that the area occupied at a specific time is less than the total area required for the species to persist [30]. The discovery and incorporation of new leks in models could improve the accuracy of research at fine spatial scales and help guide managers in their conservation efforts [17]. Leks are often central to the annual range for nonmigratory sage-grouse populations [3]. Previous research by Coates et al. (2013) [8] demonstrated the utility of lek-centered management in Nevada, where protected areas centered at leks with radii of 7.5 km included almost all seasonal space use for most populations. The success of that approach depends on managers knowing the location of nearly all leks in a given area [8] — which our predictions directly support.

Management Implications

These data can be useful for rangeland managers in the state of Nevada seeking to thoroughly understand and improve sage-grouse population estimates through lek locations. Our study could lead to the discovery

of unknown leks, improving estimates, projections, and connectivity analyses for Nevada populations. Our findings can help direct limited resources to the areas that would benefit most, by delineating the most suitable lek locations across the state. Because the analysis is a scripted, openly licensed pipeline operating entirely on public environmental data (lek locations by agency agreement), it can be re-run as new LANDFIRE, 3DEP, and lek-status data are released, and applied directly to other populations of sage-grouse across their geographic range where large lek datasets exist.

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References [1]–[30] as in the 2024 draft (Knick 2003; Crawford 2004; Connelly 2011; Coates 2013; Kirol 2015; Van Horne 1983; WAFWA 2015; Knick 2013; Coates 2014; Coates 2016; Ricca 2018; O'Donnell 2019; Coates 2020; Coates 2021; Manly 2007; Aldridge & Boyce 2007; Aldridge 2008; Carpenter 2010; Fedy 2014; Fedy 2015; Baxter 2017; LANDFIRE; R Core Team; Mortelliti 2010), with the following updates and additions:

- [21] *removed* (ESRI ArcGIS Pro no longer used; processing described in Methods is implemented in GDAL, Python, and R — pipeline archived at github.com/ryanjshields/nv-sagegrouse).
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Supporting Information

S1 Table. Comparison of the available-sample mean and the study-area mean of each continuous variable used in the RSF.

Variable	Study-area mean	Available-sample mean	85% CI low	85% CI high	Covered
Elevation (m)	1,878	1880	1875	1885	yes
Slope (°)	8.219	8.194	8.051	8.337	yes
Ruggedness (VRM)	0.002068	0.001972	0.001895	0.002049	no
Curvature	−0.001529	0.0004905	−0.004819	0.0058	yes
Distance to any road (m)	—	893.1	874.9	911.3	yes
Distance to paved road (m)	—	12550	12370	12730	yes
Distance to unpaved road (m)	—	914.1	895.7	932.6	yes

S2 Table. Continuous Boyce evaluation: predicted-to-expected (P/E) ratio of lek occurrence by predicted-probability class. P/E > 1 indicates a class captures more leks than its areal share of the study region alone would predict; a monotone rise in P/E across classes yields an index near 1 [42].

Predicted probability (class midpoint)	P/E ratio (lek share / area share)
0.059	0.36
0.111	1.23
0.164	2.05
0.216	2.8
0.269	3.71
0.321	4.29
0.374	5.03
0.426	4.46
0.479	4.21
0.531	6.25

Continuous Boyce index: 0.915

P/E is the predicted-to-expected ratio: the proportion of the used leks whose predicted probability falls in a class, divided by the proportion of the study area in that class. $P/E = 1$ means leks occur in the class at exactly the rate its area alone would predict; $P/E > 1$ means the class captures more leks than its share of the landscape. The continuous Boyce index (Hirzel et al. 2006) is the Spearman rank correlation between P/E and class midpoint: it ranges from -1 to 1 , where values near 1 indicate lek density rises monotonically with predicted suitability, 0 indicates a surface no better than chance, and negative values indicate counter-prediction.